1.

Exercise 4

Define an RNN for regression, having 2 inputs, a hidden state of dimension 3, and 2 outputs. The activation function for the hidden state is ReLU. Write the equations for two time steps of forward pass through the network. For the inputs $\boldsymbol{X}_1 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$, $\boldsymbol{X}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, if the weight matrices and bias vector are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the outputs $\boldsymbol{O}_1$ and $\boldsymbol{O}_2$ of the network. Specify the total number of real parameters (weights and biases) for the layer.

Solution:

we have the following formulas:

$$\boldsymbol{H}_t = \phi(\boldsymbol{X}_t \boldsymbol{W}_{xh} + \boldsymbol{H}_{t-1} \boldsymbol{W}_{hh} + \boldsymbol{b}_h),$$

$$\boldsymbol{O}_t = \boldsymbol{H}_t \boldsymbol{W}_{hq} + \boldsymbol{b}_q,$$

where $\boldsymbol{X}_t \in \mathbb{R}^{n \times d}$, $\boldsymbol{W}_{xh} \in \mathbb{R}^{d \times h}$, $\boldsymbol{W}_{hh} \in \mathbb{R}^{h \times h}$, $\boldsymbol{b}_h \in \mathbb{R}^{1 \times h}$, $\boldsymbol{H}_t \in \mathbb{R}^{n \times h}$, $\boldsymbol{W}_{hq} \in \mathbb{R}^{h \times q}$, $\boldsymbol{b}_q \in \mathbb{R}^{1 \times q}$, $\boldsymbol{O}_t \in \mathbb{R}^{n \times q}$, and

$$\phi(x) = \text{ReLU}(x)$$

in our case, $n = 2$, $d = 2$, $h = 3$, $q = 2$

we take

$$\boldsymbol{W}_{xh} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3},$$

$$\boldsymbol{W}_{hh} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 3},$$

$$\boldsymbol{b}_h = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \in \mathbb{R}^{1 \times 3},$$

$$\boldsymbol{W}_{hq} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2},$$

$$\boldsymbol{b}_q = \begin{bmatrix} 1 & -1 \end{bmatrix} \in \mathbb{R}^{1 \times 2},$$

$$\boldsymbol{H}_0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$$

- now, we apply the formulas:

$$\begin{aligned} \boldsymbol{H}_1 &= \phi\left(\begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 2 & 0 & 0 \\ 2 & -1 & 0 \end{bmatrix}\right) \\ &= \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3}, \end{aligned}$$

$$\begin{aligned} \boldsymbol{H}_2 &= \phi\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 & -2 \\ 2 & 0 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 3 & 1 & -3 \\ 4 & 0 & -2 \end{bmatrix}\right) \\ &= \begin{bmatrix} 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3}, \end{aligned}$$

H_1 W_{hq} b_q

$t=1$

$$
\begin{aligned}
o_1 &= \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

H_2 W_{hq} b_q

$t=2$

$$
\begin{aligned}
o_2 &= \begin{bmatrix} 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 1 \\ 4 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & 0 \\ 5 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

$$
\begin{aligned}
\text{Total parameters} &= 2\times 3 + 3\times 3 + 1\times 3 + 3\times 2 + 1\times 2 \\
&= 6+9+3+6+2 \\
&= 26.
\end{aligned}
$$

W_{xh} W_{hh} b_h W_{hq} b_q

$$\text{param} = d\times h + h\times h + 1\times h + h\times q + 1\times q = 26$$

(2.) Ex de la curs S11

GRU

$$x_1 = \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 2} \qquad x_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 2}$$

$h = 3$, $d = 2$ (coloane), $n = 2$ (nr de ese) (linii), $T = 2$

$d\times h$

$$W_{xh} = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 3}$$

$d\times h$

$$W_{xz} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 3}$$

$h\times h$

$$W_{hr} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3}$$

$$W_{hz} = \begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3} \quad (h \times h)$$

$$b_h = \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} \in \mathbb{R}^{1\times 3} \quad (1 \times h)$$

$$b_z = \begin{pmatrix} 0 & -1 & 1 \end{pmatrix} \in \mathbb{R}^{1\times 3} \quad (1 \times h)$$

$$W_{xh} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 3} \quad (d \times h)$$

$$W_{hh} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \in \mathbb{R}^{3\times 3} \quad (h \times h)$$

$$b_h = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{1\times 3}$$

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 3}$$

- $t = 1$

$$R_1 = \nabla\left(x_1 W_{xh} + H_0 W_{hh} + b_h\right) =$$

$$= \nabla\left(\left(\begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \right.\right.$$

$$\left.\left. + \begin{pmatrix} 1 & 0 & -1 \end{pmatrix}\right) =$$

$$= \nabla\left(\left(\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \end{pmatrix} + 0^{2\times 3} + \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}\right)\right) =$$

$$= \begin{pmatrix} \tilde{V}(2) & \tilde{V}(-1) & \tilde{V}(0) \\ \tilde{V}(2) & V(0) & \tilde{V}(-1) \end{pmatrix}$$

$$z_1 = V\left(x_1 W_{xz} + H_0 W_{hz} + b_z\right) =$$

$$= V\left(\begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix} + 0^{2\times 3}\begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -1 & 1 \end{pmatrix}\right) =$$

după calcule $= \begin{pmatrix} \tilde{V}(-1) & \tilde{V}(-1) & \tilde{V}(3) \\ \tilde{V}(0) & V(1) & V(2) \end{pmatrix}$

$$\tilde{H}_1 = \tanh\left(x_1 W_{xh} + (R_1 \circ H_0) W_{hh} + b_h\right) =$$

$$= \tanh\left(\begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \end{pmatrix} + \left(\begin{pmatrix} \tilde{V}(2) & \tilde{V}(-1) & V(0) \\ V(2) & \tilde{V}(0) & \tilde{V}(-1) \end{pmatrix} \circ 0^{2\times 3}\right) \cdot \right.$$

$$\left. \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}\right) =$$

$$= \tanh\left(\begin{pmatrix} 2 & 1 & -1 \\ 2 & 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} \tanh(2) & \tanh(1) & \tanh(-1) \\ \tanh(2) & \tanh(0) & \tanh(0) \end{pmatrix}$$

nr de param $= d\times h + d\times h + h\times h + h\times h + 1\times h + 1\times h + d\times h + h\times h + 1\times h = 6+6+9+9+3+3+6+9+3 = 54$

$$H_1 = Z_1 \odot H_0 + (1 - Z_1) \odot \tilde{H}_1 =$$

$$= \begin{pmatrix} \sigma(-1) & \sigma(-1) & \sigma(3) \\ \sigma(0) & \sigma(-1) & \sigma(2) \end{pmatrix} \odot 0^{2\times 3} + \begin{pmatrix} 1-\sigma(-1) & 1-\sigma(-1) & 1-\sigma(3) \\ 1-\sigma(0) & 1-\sigma(-1) & 1-\sigma(2) \end{pmatrix} \odot$$

$$\odot \begin{pmatrix} \tanh(2) & \tanh(1) & \tanh(-1) \\ \tanh(2) & \tanh(0) & \tanh(0) \end{pmatrix} =$$

$$0^{2\times 3} + [\dots]$$

3.

Question 8
Not yet answered
Marked out of 1.00
Flag question

Compute symbolically the total number of parameters of a GRU layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

Maximum file size: 5MB, maximum number of files: 1

$n = 2 \quad d = 3 \quad h = 4$

$x_t \in \mathbb{R}^{n\times d} = \mathbb{R}^{2\times 3}$

$H_{t-1} \in \mathbb{R}^{n\times h} = \mathbb{R}^{2\times 4}$

$R_t \in \mathbb{R}^{n\times h} = \mathbb{R}^{2\times 4} \qquad R_t = \sigma(x_t W_{xh} + H_{t-1} W_{hh} + b_h)$

$Z_t \in \mathbb{R}^{n\times h} = \mathbb{R}^{2\times 4} \qquad Z_t = \sigma(x_t W_{xz} + H_{t-1} W_{hz} + b_z)$

Parametrii

$W_{xh}, W_{xz} \in \mathbb{R}^{d\times h} = \mathbb{R}^{3\times 4} \Rightarrow 2\cdot(3\cdot 4) = 24$

$W_{hh}, W_{hz} \in \mathbb{R}^{h\times h} = \mathbb{R}^{4\times 4} \Rightarrow 2\cdot(4\cdot 4) = 32$

$b_h, b_z \in \mathbb{R}^{1\times h} = \mathbb{R}^{1\times 4} \Rightarrow 2\cdot(1\cdot 4) = 8$

$\tilde{H}_t \in \mathbb{R}^{n\times h} = \mathbb{R}^{2\times 4}$

64

$$\tilde{H}_t = \tanh\left(x_t W_{xh} + (R_t \odot H_{t-1}) W_{hh} + b_h\right)$$

Parametrii:

$W_{xh} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \Rightarrow 3 \cdot 4 = 12$

$W_{hh} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \Rightarrow 4 \cdot 4 = 16$

$b_h \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \Rightarrow 1 \cdot 4 = 4$

32

$64 + 32 = 96$

4. Compute symbolically the total number of parameters of an LSTM layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: n=2, d=3, h=4.

Compute symbolically the total number of parameters of an LSTM layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

$n = 2 \quad d = 3 \quad h = 4$

$x_t \in \mathbb{R}^{n \times d} = \mathbb{R}^{2 \times 3}$

$H_{t-1} \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4}$

$I_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \qquad I_t = \sigma(x_t W_{xi} + H_{t-1} W_{hi} + b_i)$

$F_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \qquad F_t = \sigma(x_t W_{xf} + H_{t-1} W_{hf} + b_f)$

$O_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \qquad O_t = \sigma(x_t W_{xo} + H_{t-1} W_{ho} + b_o)$

Param:

$W_{xi}, W_{xf}, W_{xo} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \Rightarrow 3 \cdot (4 \cdot 3) = 36$

$W_{hi}, W_{hf}, W_{ho} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \Rightarrow 3 \cdot (4 \cdot 4) = 48$

$b_i, b_f, b_o \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \Rightarrow 3 (1 \cdot 4) = 12$

96

$\tilde{C}_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4}$

$\tilde{C}_t = \tanh(x_t W_{xc} + H_{t-1} W_{hc} + b_c)$

$W_{xc} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \Rightarrow 3 \cdot 4 = 12$

$W_{hc} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \Rightarrow 4 \cdot 4 = 16$

$b_c \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \Rightarrow 1 \cdot 4 = 4$

32

param = $96 + 32 = 128$

4. Draw the diagram of an RNN for regression, having 3 inputs, a hidden state of dimension 2, and 3 outputs. The activation function for the hidden state is ReLU. For the inputs $\boldsymbol{X}_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $\boldsymbol{X}_2 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$, if the weight matrices and bias vector are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the outputs $\boldsymbol{O}_1$ and $\boldsymbol{O}_2$ of the network. Specify the total number of real parameters (weights and biases) for the network.

$n = 2$, $d = 3$, $h = 2$, $q = 3$

d: nr inputuri

Solution:
we have the following formulas:

$$\boldsymbol{H}_t = \phi(\boldsymbol{X}_t \boldsymbol{W}_{xh} + \boldsymbol{H}_{t-1} \boldsymbol{W}_{hh} + \boldsymbol{b}_h),$$

$$\boldsymbol{O}_t = \boldsymbol{H}_t \boldsymbol{W}_{hq} + \boldsymbol{b}_q,$$

where $\boldsymbol{X}_t \in \mathbb{R}^{n \times d}$, $\boldsymbol{W}_{xh} \in \mathbb{R}^{d \times h}$, $\boldsymbol{W}_{hh} \in \mathbb{R}^{h \times h}$, $\boldsymbol{b}_h \in \mathbb{R}^{1 \times h}$, $\boldsymbol{H}_t \in \mathbb{R}^{n \times h}$, $\boldsymbol{W}_{hq} \in \mathbb{R}^{h \times q}$, $\boldsymbol{b}_q \in \mathbb{R}^{1 \times q}$, $\boldsymbol{O}_t \in \mathbb{R}^{n \times q}$, and

$$\phi(x) = \text{ReLU}(x)$$

W_{xh} (3×2) $= \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$ W_{hh} (2×2) $= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ W_{hq} (2×3) $= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b_h (1×2) $= (1 \quad 0)$ b_q (1×3) $= (1 \quad 0 \quad 0)$

H_0 (2×2) $= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$H_1 = \tilde{V}\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} + O^{2\times 2} W_{hh} + (1 \quad 0) \right) =$

$= \tilde{V}_{\text{relu}}\left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$

$$O_1 = H_1 W_{hq} + b_q =$$

$$= \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

→ la fel și pt H_2, O_2

$$\text{param} = 3 \times 2 + 2 \times 2 + 1 \times 2 + 2 \times 3 + 1 \times 3 =$$

$$= 6 + 4 + 2 + 6 + 3 = 21$$

8. Write the equations for the forward pass of an RNN with two hidden states. Compute symbolically the total number of parameters (weights and biases) of an RNN with two hidden states, knowing that the number of examples is n, the number of inputs is d, the number of hidden units of both hidden states is h, and the number of outputs is q. Numerical application: $n = 2, d = 3, h = 4, q = 2$. For the numerical application, write one step through the RNN, with the input, weights, and biases generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros).

$l = 2$ (nr. of hidden layers)

$n = 2$, $d = 3$, $h = 4$, $q = 2$ $\quad X_i \in \mathbb{R}^{2 \times 3}$

$X_t \in \mathbb{R}^{n \times d} = \mathbb{R}^{2 \times 3}$

$H_t^{(l)} \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4}$

$Ot \in \mathbb{R}^{n \times q} = \mathbb{R}^{2 \times 2}$

$H_t^{(0)} = X_t \Rightarrow H_t^{(l)} = \phi\left(H_t^{(l-1)} W_{xh}^{(l)} + H_{t-1}^{(l)} W_{hh}^{(l)} + b_h^{(l)} \right)$

$W_{xh}^{(l)} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \longrightarrow 2\,(3 \cdot 4) = 2 \cdot 12 = 24$

$W_{hh}^{(l)} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \longrightarrow 2\,(4 \cdot 4) = 2 \cdot 16 = 32$

$b_h^{(l)} \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \longrightarrow 2 \cdot (1 \cdot 4) = 2 \cdot 4 = 8$

64

$O_t = H_t^{(L)} W_{hq} + b_q$

$W_{hq} \in \mathbb{R}^{h \times q} = \mathbb{R}^{4 \times 2} \longrightarrow 4 \cdot 2 = 8$

$b_q \in \mathbb{R}^{1 \times q} = \mathbb{R}^{1 \times 2} \longrightarrow 1 \cdot 2 = 2$

10

$\Rightarrow$ param $= 64 + 10 = 74$

$X_i \in \mathbb{R}^{2 \times 3} \longrightarrow$ luăm unul singur (un pas)

$$X_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$W_{hh}^{(1)} = I_4 \qquad b_h^{(1)} = (1 \;\; 0 \;\; 0 \;\; 0)$

$W_{hh}^{(2)} = I_4 \qquad b_h^{(2)} = (-1 \;\; 0 \;\; 0 \;\; 0)$

$$W_{xh}^{(1)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$W_{xh}^{(2)} = I_4$

$$W_{hq} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \qquad b_q = (1 \;\; 0)$$

ReLu

$H_t^{(0)} = X_1 \qquad H_0^{(1)} = 0^{2 \times 4} \qquad H_0^{(2)} = 0^{2 \times 4}$

$$H_1^{(1)} = \text{ReLu}\left(X_1 \cdot W_{xh}^{(1)} + H_0^{(1)} W_{hh}^{(1)} + b_h^{(1)} \right) =$$

$$= \text{ReLu}\left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + 0^{2\times 4} + \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \right) =$$

$$= \text{ReLu}\left(\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \right) =$$

$$= \begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$H_1^{(2)} = \text{ReLu}\left(H_1^{(1)} W_{xh}^{(2)} + H_0^{(2)} W_{hh}^{(2)} + b_h^{(2)} \right) =$$

$$= \text{ReLu}\left(\begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + 0_{2\times 4} \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 & 0 & 0 \end{pmatrix} \right) =$$

$$= \text{ReLu}\left(\begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \right) =$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$O_1 = H_1^{(2)} W_{hq} + b_q =$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$